Pranav Chopra

Personal Portfolio: thepranavchopra.com
437-213-4288 pranav.chopra06@gmail.com linkedin.com/in/pchopr2 github.com/prananchh

Education

University of Western Ontario **Sep 2023 – Apr 2027**
Bachelor of Software Engineering, Co-op *London, Ontario*

Skills

Programming: Dart, Kotlin, Flutter, Python, Java, C, HTML5/CSS3, JavaScript, C++, SQL, Node.js, Arduino, LaTeX
Software: Data Structures & Algorithms, Cloud (Azure), Version Control, VS Code, AWS, Generative AI, Predictive Analytics, Data Handling, Databases
Technologies/Frameworks: .NET, Power Platform, Git/GitHub, CAD, MS Office, Computer Vision, React, Django, Flask, OpenCV, MediaPipe, Agile, Figma, Miro, Adobe CC, Photoshop, AutoCAD, Google Analytics, Unity

Technical Experience

AI/ML Engineering Intern – Canadian Musicians Co-op **Jun 2025 – Present**

- Built and integrated an **audio-input classification service** into the web backend for **real-time singing feedback** (pitch, timing, vibrato/clarity).
- Designed and trained a **CNN** in **TensorFlow/Keras** with end-to-end audio preprocessing and efficient batching.
- Achieved **99% validation accuracy** and **98% test accuracy** on a held-out set, demonstrating strong generalization.
- Prepared rollout to **300+ musicians** at the Canadian Musicians Co-operative for scalable, real-time coaching.

Machine Learning Engineer – Western AI **Sep 2025 – Present**

- Engineered a **multimodal fusion pipeline** integrating **video (CLIP)**, **audio (OpenL3)**, and **text (Sentence Transformers)** embeddings from **10K+ short-form videos**; implemented a transformer-based attention network achieving **78% AUROC** for **viral content prediction** (>1M views).
- Built a dimensionality reduction framework using **PCA**, **NMF**, and **autoencoders** to derive **15 latent virality factors** from **512-d embeddings**, reducing **compute by 65%** while preserving **92% variance**.
- Trained **MLP regression** and **pairwise ranking networks** for **view-count trajectory forecasting** (**Spearman ρ = 0.71**, **RMSE = 145K**), enabling data-driven pre-release optimization.
- Worked with a team of 5 engineers to deploy an end-to-end ML pipeline with spectral clustering, **t-SNE**, and **UMAP** visualizations, identifying **8 virality clusters** that increased engagement by **34%**.

Project Manager – Western Cyber Society **Oct 2025 – Present**

- Leading a team of 5 engineers to build **LeakSight**, an ML-driven cybersecurity platform analyzing timing, power, and signal data to uncover side-channel vulnerabilities.
- Architected **data pipelines** and **model training workflows** using Python, NumPy, SciPy, scikit-learn, and PyTorch, containerized with Docker for reproducibility and security.
- Delivered a **real-time leak detection system** with **92%+** accuracy and an interactive Streamlit dashboard showcasing effective countermeasures for data exposure.

Projects

LearnBridge – AI Agent for Students **Mar 2025**

- Architected **React-based multi-agent workflows** with **memory**, **parallel execution**, and **tool integration** (Google Classroom & Calendar), with prompt-tuned orchestration in a **Streamlit UI**.
- Accelerated assignment onboarding by extracting **key concepts** from announcements/slides; generated **formatted summaries** (graphs, tables) and auto-added deadlines to Google Calendar (**80% faster** content access).

Automated Bionic Arm – Hack the Heart (University of Toronto) **Oct 2024**

- Prototyped a **low-cost bionic arm** using **Python/OpenCV** for **real-time gesture recognition** (**95% accuracy**); implemented **modular OOP control** for responsive actuation.
- Integrated **computer vision pipelines** for hand tracking (**sub-100ms latency**); coordinated **150+ GitHub commits**; received **finalist recognition** demonstrating accessible **AI-powered prosthetics**.